

How Terraform, Pulumi and OpenTofu compare with each other

Feature	Terraform	Pulumi	OpenTofu
Language support	HCL	Python, TS, Go	HCL
Flexibility	Moderate	High	Moderate
Ecosystem	Very large	Growing	Growing
Open source	No (BSL)	(Partial)	Fully open
Enterprise-ready	Yes	Yes	In progress
Best for	DevOps teams	Developers	Open source advocates

- **Choose Pulumi** if you're a developer-first team looking to build infrastructure with the power of familiar languages.
- **Choose OpenTofu** if you want the freedom of open source with Terraform compatibility and community-driven innovation. **END** 🐧

Final recommendation

The right IaC tool depends on your team's structure, values, and technical needs:

- **Choose Terraform** if you want a stable and mature tool backed by years of adoption (and you're okay with BSL restrictions).

By: Rajnesh Devi

The author is assistant professor at the Yamuna Group of Institutions.

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The trend towards unified processing engines: Unified processing engines, such as Apache Beam, Flink and Delta Lake, enable developers to manage batch and stream workloads through one API or execution plan. This trend removes the requirement for separate systems and makes data engineering easier. Developers can write once and run oriented code anywhere (on real time or historical data).

The emergence of serverless and edge streaming: Cloud-native and serverless architectures that allow data engineers to run pipelines without worrying about infrastructure are becoming the norm. AWS Kinesis, Azure Stream Analytics and Google Cloud Dataflow provide data engineers the opportunity to think about code and not servers. Additionally, edge streaming is emerging in industries with a high amount of IoT data, where data can be processed when and where it's created so that latency and bandwidth costs can be reduced.

Auto-generated pipelines with AI: AI is now encroaching the domain of data engineering. New tools are emerging to automatically generate, optimise, and monitor pipelines via schema, usage/pattern, or natural language. AI

pipelines reduce human error, speed up a team's development, and allow teams to reconfigure pipelines in the moment.

With stream engines offering replay features and batch engines enabling near real-time capability, the lines are fading between stream and batch processing. The result is an avenue for processing and not a separation of two concepts, where data flows fluidly across historic and live contexts allowing businesses to act with speed and nuance. Batch or stream, each has a purpose; the true power comes from their combination in hybrid architectures that facilitate both speed and depth.

The democratisation of data integration through open source tools has put advanced data integration

capabilities in the hands of everyone — from startups to Fortune 500 companies. Kafka, Flink, Spark, and Beam are examples of the technology underpinning the way organisations are acquiring, ingesting, processing, and acting on data in real-time. Unified processing engines and serverless platforms are reducing operational and project effort, enabling less friction in development.

With the line between stream and batch processing blurring, the future of data integration will be about flexibility and intelligence. As AI automation, edge processing, and intelligent orchestration evolve and create networks that provide real-time pipelines and scalability, the limitations of both streams and batches may be overcome soon. **END** 🐧

By: Dibyendu Banerjee and Sourav Kairi

Dibyendu Banerjee has 20+ years of experience in the IT industry, specialising in artificial intelligence and machine learning. He is a certified data scientist, certified deep learning and NLP practitioner, and generative AI specialist.

Sourav Kairi is actively contributing to various AI and ML projects at a Tier 1 IT company. He is also a member of the Board of Studies (BOS) at a well-known engineering college. This reflects his dedication to both tech innovation and academic excellence.

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